



IndianOil

JOURNEY RISK MANAGEMENT (JRM) STUDY

Salem Terminal to K.M.B. FUELS AND OILS

Objective of the JRM Report

This JRM report is designed to ensure compliance with the Central Motor Vehicle Rules, 1989 (CMVR), AIS 140 standards, and the Road Transport Safety Policy (RTSP). It provides a comprehensive risk assessment for the transportation of hazardous materials along specified routes. By integrating these legal frameworks, the report offers a broad strategy for identifying and mitigating route-specific risks.

Regulatory Compliance

The report complies with the Central Motor Vehicles (Eleventh Amendment) Rules, 2022, mandating safe transportation practices for N2 and N3 category vehicles carrying hazardous materials. These rules require detailed route assessments, especially regarding road conditions, speed limits, and risk areas, to ensure safety compliance.

Risk Management Strategy

This report categorizes transportation routes into high-risk and medium-risk areas, with a focus on factors such as sharp turns, accident-prone regions, and elevation changes. The goal is to provide actionable recommendations to minimize these risks, including speed regulations, driver warnings for hazardous zones, and the option of alternate routes.

Compliance with the Road Transport Safety Policy (RTSP)

The report integrates RTSP provisions, including mandatory driving hours, rest periods, and nighttime driving restrictions. It ensures that drivers follow official guidelines, such as taking prescribed rest breaks and avoiding dangerous road conditions like poor visibility, heavy crowds, or high-traffic areas during peak hours.

Emergency Preparedness and Response

The report highlights the significance of predetermined emergency stops for refueling, rest, and overnight stays. It includes protocols for safe responses to road hazards, alternative routes, and rerouting processes if roads are closed or severe weather arises. This aligns with the RTSP emphasis on driver safety and rapid emergency response.

Environmental Considerations

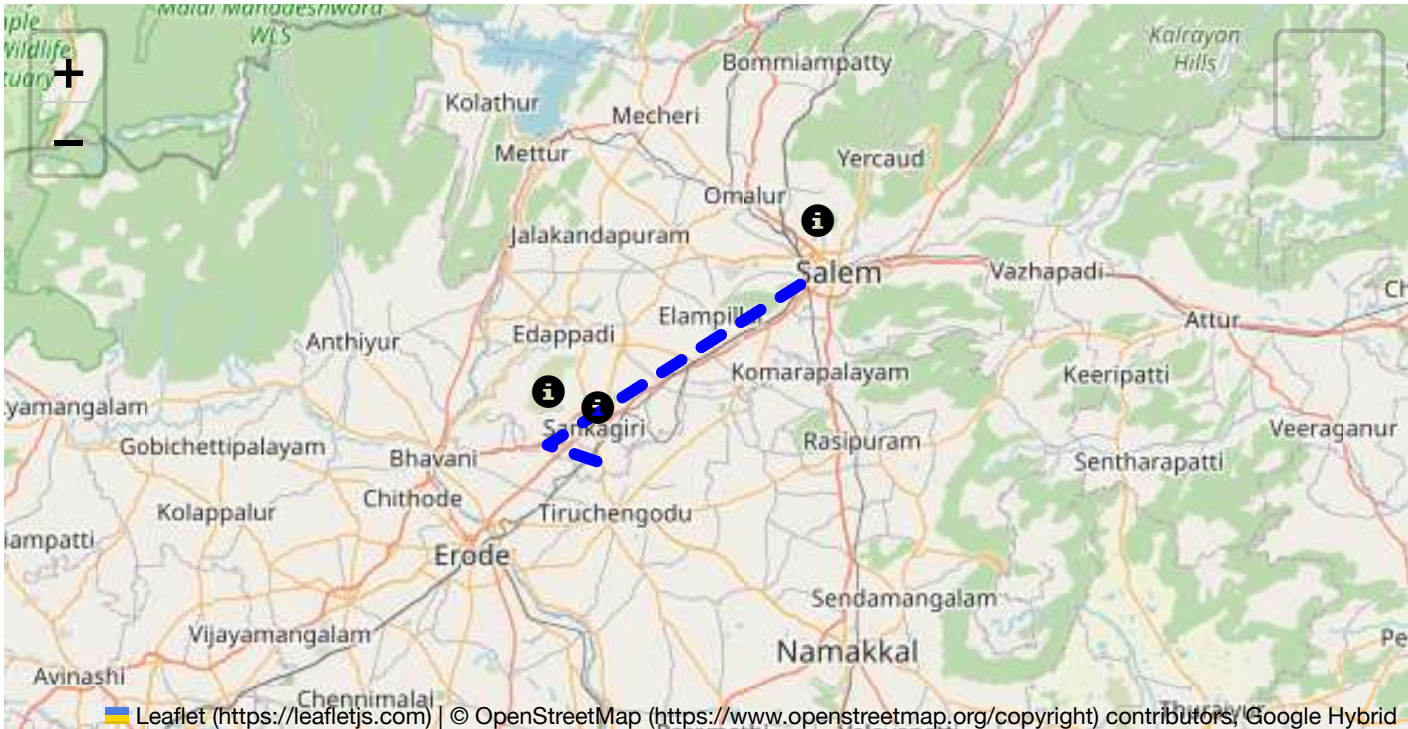
The JRM report addresses environmental risks along the route, ensuring compliance with environmental protection laws in ecologically sensitive zones. It suggests strategies such as identifying areas near water bodies, forests, or populated regions and implementing safety measures to minimize environmental impacts during transport.

Journey Risk Mitigation

The report includes route-specific risk assessments, detailed journey charts, and defensive driving guidelines for each transport route. Integration with vehicle tracking systems guarantees real-time warnings on hazardous areas, speed limits, and mandatory stops, consistent with RTSP and CMVR safety norms.

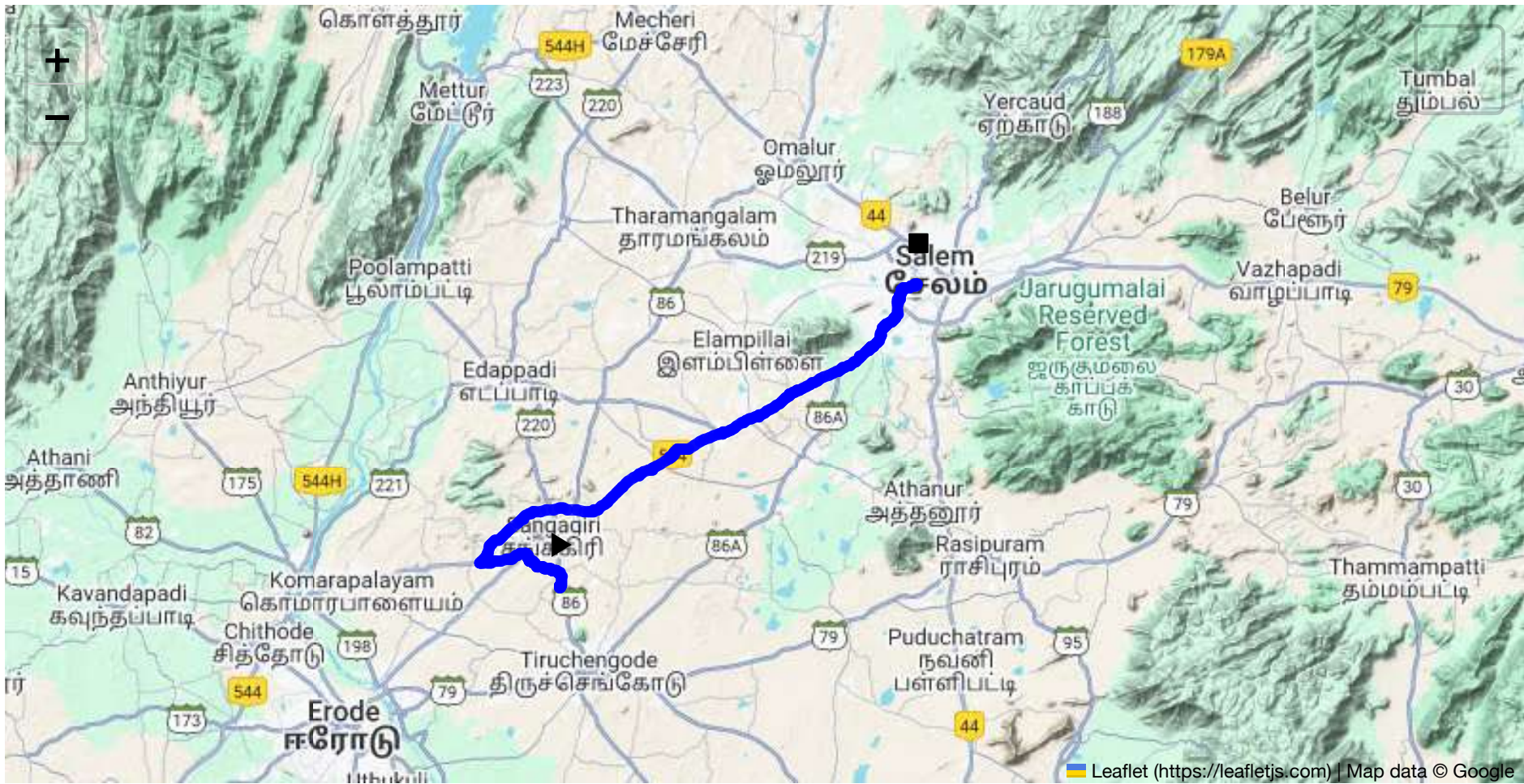
Compliance with Government Directives

This report fully adheres to governmental directives regarding hazardous material transportation, implementing mandatory speed limits, nighttime driving restrictions, and comprehensive driver briefings and real-time alerts about route-related risks.



Route Summary:
Total Distance: 51.95 km
Estimated Duration: 1.0 hours
Adjusted Duration (Heavy Vehicle): 1.3 hours

Start: (11.4381, 77.8734)
End: (11.652495, 78.131729)



Welcome to the Journey Risk Management Study

Route Safety Analysis Report

1. Overview of the Route Map

The route consists mainly of rural roads and small highways, covering approximately 51.95 kilometers from CVQF+23W, Sangagiri to 413, Mollapillayar Kovil Rd, Sairaj Nagar, Pallapatti in Salem, Tamil Nadu. It traverses through FR48+7XQ, Bhavani Main Road, Padaiveedu, providing a direct path for traffic between these points.

2. Typical Weather Conditions and Potential Weather-Related Hazards

Tamil Nadu experiences a tropical climate with significant temperature fluctuations and seasonal monsoon rains. The route is particularly vulnerable to heavy rainfall from June to September, which can cause flooding and poor visibility. Monsoons often lead to slippery roads and increased risk of hydroplaning. Therefore, drivers should be cautious during these months, especially in low-lying areas.

3. Analysis of Traffic Patterns

The route does not typically experience high congestion. However, the road can see increased vehicle density during peak commuting hours in the morning from 8:00 AM to 10:00 AM and in the evening from 5:00 PM to 7:00 PM. The vicinity of Sangagiri and Salem are more prone to congestion due to local traffic.

4. Assessment of Road Quality and Infrastructure

The road infrastructure along this route varies. While main highways are generally well-maintained, some rural segments may have uneven surfaces and potholes, which can be hazardous for heavy vehicles. Road signs and markings are often less visible, particularly at night or during adverse weather conditions.

5. Suggestions for Alternative Routes for Emergencies

In case of road blockages or emergencies, alternate routes include using NH44 from Sankagiri towards Salem, which is better maintained and wider, although it could be longer. Local knowledge of smaller connecting roads may provide additional detour options.

6. Summary of Local Regulations Affecting Hazardous Material Transport

Tamil Nadu regulations mandate strict adherence to specific routes for hazardous materials, and transport should be informed by local guidelines to avoid routes passing through densely populated areas. Permits and proper vehicle labeling are required.

7. Overview of Historical Incidents Involving Heavy Vehicles or Hazardous Materials

There have been a few incidents involving heavy vehicles in the region, often attributable to road quality issues or driver error. No significant incidents exclusive to hazardous materials have been reported, indicating moderate safety compliance.

8. Environmental Considerations and Sensitive Areas

The route goes through several agricultural zones. Care is needed to prevent spills or accidents that could impact local farmlands. Be aware of wildlife crossings, especially in rural stretches.

9. Analysis of Communication Coverage

The majority of the route enjoys decent mobile phone coverage, though there may be dead zones, especially in rural areas. Communication devices should be tested beforehand, and offline maps or GPS systems can be used as backups.

10. Estimated Emergency Response Times for Different Route Segments

Emergency response times are generally faster near urban areas like Sangagiri and Salem, typically ranging from 10 to 20 minutes. Rural areas might experience longer response times, potentially up to 30 minutes or more.

11. Overall Summary of Risk Assessment

Overall, the route poses moderate risks for transporting hazardous materials due to variable road quality, seasonal weather conditions, and traffic congestion near urban areas. Proper planning, including awareness of alternative routes and adherence to traffic and weather advisories, can minimize these risks effectively.

Safe driving practices and a comprehensive understanding of the route's challenges will enhance the safety of transport operations along this corridor. Regular updates to this analysis are recommended, especially with changing weather conditions or road infrastructure developments.

Risk Assessment - Turns

	Risk Type	Risk Level	Coordinates	Speed Limit
0	Turn	High	11.43811, 77.87348	15 KM/Hr
1	Turn	Medium	11.43956, 77.87340	30 KM/Hr
2	Turn	High	11.43968, 77.87345	15 KM/Hr
3	Turn	High	11.44029, 77.87544	15 KM/Hr
4	Turn	Medium	11.45353, 77.85697	30 KM/Hr
5	Turn	Medium	11.45839, 77.85143	30 KM/Hr
6	Turn	Medium	11.46307, 77.84941	30 KM/Hr
7	Turn	Medium	11.46316, 77.84921	30 KM/Hr
8	Turn	High	11.45542, 77.81725	15 KM/Hr
9	Blind Spot	Blind Spot	11.45631, 77.81668	10 KM/Hr
10	Turn	Medium	11.64510, 78.12033	30 KM/Hr
11	Turn	Medium	11.64518, 78.12029	30 KM/Hr
12	Turn	High	11.64970, 78.12064	15 KM/Hr
13	Turn	Medium	11.64986, 78.12096	30 KM/Hr
14	Turn	Medium	11.65102, 78.12280	30 KM/Hr
15	Blind Spot	Blind Spot	11.65293, 78.13182	10 KM/Hr

Emergency Locations

	type	name	coordinates	speed_limit	risk_level
2	hospital	C. N. Hospital	11.55724, 78.0096	30 km/h	Medium
3	hospital	Annapoorna Medical College Hospital	11.57693, 78.037627	30 km/h	Medium
4	hospital	Vinayaka Mission's Homeopathic Medical College & Hospital	11.583617, 78.059145	30 km/h	Medium
5	hospital	Vinayaka mission Kirubananadah variyar medical college hospital	11.584984, 78.0618417	30 km/h	Medium
6	hospital	vims	11.5854239, 78.0639188	30 km/h	Medium
9	hospital	Vinayaka Mission’s Kirupananda Variyar Medical College	11.5977669, 78.0850992	30 km/h	Medium
13	hospital	Bharathi Hospital, Salem	11.6492647, 78.1204131	30 km/h	Medium

	type	name	coordinates	speed_limit	risk_level
14	hospital	Sisu Hospital Inida Ltd	11.6505276, 78.1205003	30 km/h	Medium

Crowded Spots

	type	name	coordinates	speed_limit	risk_level
0	school	KRP Matric. Hr. Sec School	11.4546193, 77.8142445	30 km/h	Medium
7	university	Vinayaka Missions Vinayaka Missions Research Foundation	11.5974056, 78.0843617	30 km/h	Medium
8	school	Vinayaka Missions Annapoorani Matric School	11.5971662, 78.0839876	30 km/h	Medium
10	school	Dist Education Training	11.6133799, 78.1022409	30 km/h	Medium
11	school	DIET - Salem	11.6144158, 78.1025078	30 km/h	Medium
12	school	Diamond Rays Matric HR Sec School	11.6385629, 78.1206259	30 km/h	Medium

Route Photos of Risky Spots



Risk Type: Turn
Risk Level: High
Speed Limit: 15 KM/Hr
Coordinates: 11.44029, 77.87544



Risk Type: Turn
Risk Level: Medium
Speed Limit: 30 KM/Hr
Coordinates: 11.45353, 77.85697



Risk Type: Turn
Risk Level: Medium
Speed Limit: 30 KM/Hr
Coordinates: 11.45839, 77.85143



Risk Type: Turn
Risk Level: Medium
Speed Limit: 30 KM/Hr
Coordinates: 11.46307, 77.84941



Risk Type: Turn
Risk Level: Medium
Speed Limit: 30 KM/Hr
Coordinates: 11.46316, 77.84921



Risk Type: Turn
Risk Level: High
Speed Limit: 15 KM/Hr
Coordinates: 11.45542, 77.81725



Risk Type: Blind Spot
Risk Level: Blind Spot
Speed Limit: 10 KM/Hr
Coordinates: 11.45631, 77.81668



Risk Type: Turn
Risk Level: Medium
Speed Limit: 30 KM/Hr
Coordinates: 11.64510, 78.12033



Risk Type: Turn
Risk Level: Medium
Speed Limit: 30 KM/Hr
Coordinates: 11.64518, 78.12029



Risk Type: Turn
Risk Level: High
Speed Limit: 15 KM/Hr
Coordinates: 11.64970, 78.12064



Risk Type: Turn
Risk Level: Medium
Speed Limit: 30 KM/Hr
Coordinates: 11.64986, 78.12096



Risk Type: Turn
Risk Level: Medium
Speed Limit: 30 KM/Hr
Coordinates: 11.65102, 78.12280



Risk Type: Blind Spot
Risk Level: Blind Spot
Speed Limit: 10 KM/Hr
Coordinates: 11.65293, 78.13182

